Timing of Anterior Cruciate Ligament Surgery

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KEYWORDS

• Anterior cruciate ligament • ACL • Reconstruction • Timing • Knee ligament surgery • Knee injuries • Arthroscopy • Sports medicine

KEY POINTS

- There is considerable uncertainty surrounding the definition of early and delayed anterior cruciate ligament reconstruction (ACLR), with the definition for early surgery ranging from days to months and for delayed surgery ranging from weeks to years in the current literature.
- Modern arthroscopic surgical techniques and early rehabilitation have mitigated the risk of arthrofibrosis and postoperative stiffness with early ACLR.
- Early ACLR is shown to be equivalent or superior to delayed ACLR in terms of clinical outcomes, subjective postoperative knee function, the incidence of intra-articular injury within the timeframe between injury and surgery, as well as cost-effectiveness.
- Concomitant surgical procedures addressing anterolateral instability, anteromedial instability, and meniscus injury may be more advantageous when performed in conjunction with early ACLR.

INTRODUCTION

Anterior cruciate ligament (ACL) tears are one of the most frequent injuries to the ligamentous structures of the knee. Although arthroscopic ACL reconstruction (ACLR) is the gold standard for the surgical treatment of ACL tears, the ACL surgeon needs to be aware of the wide range of patient-related, injury-related, and surgery-related factors influencing treatment outcomes. Although the timing of ACLR is one such factor, there currently is no clear consensus about the definition of early and delayed ACLR or the optimal timing of surgery following ACL tears. Surgical treatment is generally

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recommended to restore knee stability in patients with high demand for knee-strenuous activity but the decision of when to perform ACLR varies among surgeons and may be influenced by factors such as patient age, activity level, and associated injuries.

Acute knee ligament injuries were historically treated with open repair followed by postoperative immobilization with a cast. Because early surgery often resulted in a high incidence of knee stiffness and arthrofibrosis, delayed surgical treatment was proposed as an alternative, which provides time for joint inflammation to subside, recovery in knee range of motion (ROM), and restoration of muscle strength, potentially diminishing the rate of postoperative joint stiffness. However, a greater incidence of secondary intra-articular damage and potential quadriceps atrophy were proposed as drawbacks inherent to delayed ACL surgery. As a result, the complex interplay between the risk of timing-associated complications, economic factors, surgeon preference, and patient motivation to return to preinjury activity level ultimately determines surgical timing in ACLR.

Despite the growing awareness of the role of surgical timing, consensus surrounding the definition of the cutoff time interval from injury distinguishing early and delayed ACLR is currently lacking. Cutoff time intervals ranging between 8 days and[1] 12 months[2] have been reported for early isolated ACLR, whereas a cutoff as early as 3 weeks was used for delayed isolated ACLR in a recent meta-analysis,[3] emphasizing the heterogeneity with respect to timing among existing studies.

The setting of multiligament knee injury (MLKI) may also influence the decision of the operating surgeon to opt for early or delayed ACLR. Although several studies investigated outcomes following ACLR combined with the surgical or nonsurgical treatment of concomitant knee ligament injuries, evidence is currently inconclusive regarding the optimal timing of surgical treatment of MLKIs where ACLR is performed.[4]

The aim of this review is to critically assess the current literature regarding the impact of surgical timing in the setting of ACL injury treatment on a broad range of postoperative outcomes. A thorough analysis of the inconsistencies surrounding the definition of early and delayed ACLR, the influence of early versus delayed surgery on the prevalence of associated knee injuries and the influence of surgical timing on functional and clinical patient outcomes may facilitate evidence-based decision-making for surgeons performing isolated ACLR and ACLR in the setting of MLKI (**Fig. 1**).

A RATIONALE FOR THE NEED OF CLEARER DEFINITIONS OF EARLY AND DELAYED ANTERIOR CRUCIATE LIGAMENT RECONSTRUCTION

During the recent years, several systematic reviews and meta-analyses attempted to clarify the association between ACLR timing relative to the incidence of ACL injury and postoperative outcomes.[3,5–10] Although most of these studies did not determine clear differences in outcomes related to early or delayed surgery, there is unanimous agreement that conclusions are hampered by the heterogeneity of existing studies concerning the definition of early and delayed surgical timing. One systematic review considering the definitions of acute and chronic orthopedic conditions determined early ACL tears to be less than 6 week old with respect to the time of injury, whereas chronic ACL tears were defined as more than 6 month old relative to the time of injury. However, a large proportion of studies in the existing literature define ACLR as delayed even when performed before 6 weeks of ACL injury, which decreases the reliability of the currently available evidence.[10,11] Furthermore, the role of the time interval between 6 weeks and 6 months relative to the incidence of ACL injury remains unaddressed and the current definition of early and delayed ACLR remains arbitrary rather than explicit. Ain improved understanding of the inflammatory state of the knee joint in

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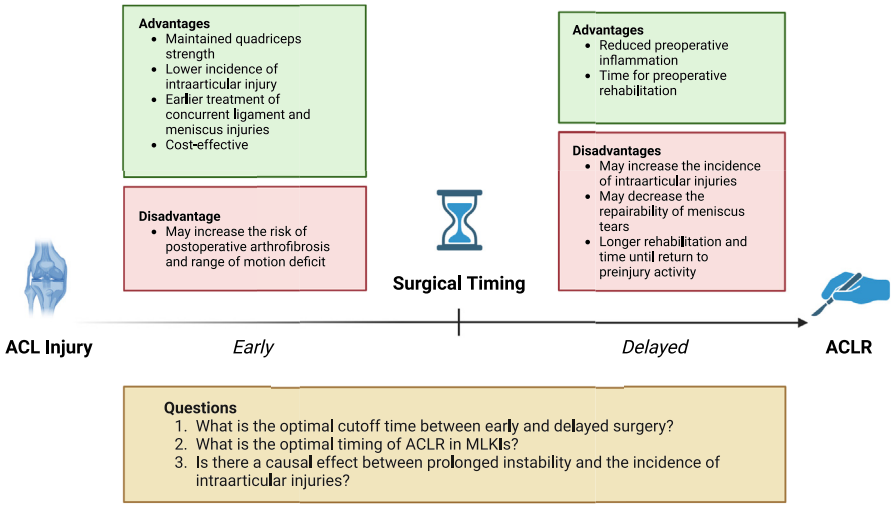

Fig. 1. The potential advantages and disadvantages of early versus delayed ACLR. ACLR, anterior cruciate ligament reconstruction; MLKI, multiple knee ligament injury.

the setting of ACL injury was also proposed to help define early and delayed surgical timing for ACLR. Synovial proinflammatory cytokine levels have been shown to increase within 24 to 72 hours of ACL injury, followed by decreasing levels during a 1-month interval before surgery.[12] However, the clinical utility of synovial biomarkers in determining the timing of ACLR remains unclear.

Despite the lack of clarity surrounding early and delayed ACLR, a recent study from the Swedish National Knee Ligament Registry determined that patients with early ACLR (<1 year from injury) reported a superior rate of acceptable knee-related symptom state and improved overall knee function compared with patients initial nonoperative treatment followed by delayed ACLR.[13] However, the time from injury to surgery was on average 6 months (standard deviation = 2.4 months) in the early ACLR group (n = 20,352) with the appropriate threshold for early ACLR remaining inconclusive.[13] In a recent systematic review and meta-analysis, comparison of 5 randomized controlled trials (RCTs) determined that patients with early isolated ACLR were on average treated 44 days earlier compared with patients who underwent delayed ACLR, with no difference in postoperative outcomes.[10] This broad time interval makes it difficult to define a clear cutoff for early and delayed ACLR and suggests that the context of surgical timing should be considered on a spectrum of months rather than a few weeks to show an impact on postoperative outcomes. Importantly, there is a lack of RCTs assessing the role of ACLR timing in the context of surgically treated MLKIs. One recent multicenter RCT reported that a complete tear of the ACL was involved in 56.5% of MLKIs and that the average time from injury to primary multiligament knee surgery was 64 days (interquartile range = 29–190 days).[4] Consequently, further high-level studies are required to determine the impact of ACLR timing in the setting of both surgically treated isolated ACL injuries and MLKIs.

THE INCIDENCE OF POSTOPERATIVE STIFFNESS AND ARTHROFIBROSIS WITH ANTERIOR CRUCIATE LIGAMENT RECONSTRUCTION TIMING

Arthrofibrosis of the knee joint is caused by excessive scar tissue formation, resulting in pain and limited potential for the patient to regain the terminal 5° of extension

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throughout knee ROM. The timing of ACL surgery has been reported to have a crucial effect on the risk of developing postoperative arthrofibrosis. A study performed in 1991 found that open ACLR performed within a week of acute ACL injury resulted in an increased rate of postoperative stiffness compared with ACLR performed at least 3 weeks from the time of acute ACL injury.[14] In the same study, delayed surgery and accelerated rehabilitation (uncommon at the time) resulted in an earlier return of strength and a significantly lower incidence of arthrofibrosis. However, delayed ACLR has been associated with an increased risk of quadriceps atrophy, which may prolong postoperative rehabilitation and may negatively influence postoperative knee function.[15]

With the advancement of arthroscopic knee surgery as the gold standard for the surgical treatment of ACL injury, the prevalence of arthrofibrosis and postoperative stiffness showed a diminishing tendency in patients undergoing ACLR, most likely due to early rehabilitation programs and improved surgical techniques.[6] More recent studies failed to demonstrate increased rates of postoperative stiffness, arthrofibrosis, and ROM deficits with a variety of cutoff thresholds between early and delayed ACLR (<2 weeks, <3 weeks, <10 weeks and beyond).[16] One RCT found that early (<8 days from injury to surgery) ACLR resulted in no ROM deficit at the 6-month or 24-month follow-up compared with delayed (6–10 weeks from injury to surgery) ACLR.[1,17] Additionally, early (<8 days from injury to surgery) surgery resulted in superior isokinetic knee flexion strength at the 24-month follow-up compared with delayed (6–10 weeks from injury to surgery) treatment.[17] Furthermore, a case series of early patellar tendon ACLRs (n = 25) performed 9 days or lesser from injury to surgery did not identify any patients with a postoperative knee extension deficit greater than 3° and knee flexion deficit greater than 5° compared with the contralateral knee at the final follow-up.[18] Authors of the referenced studies highlight the prominent role of modern arthroscopic techniques and accelerated rehabilitation protocols to facilitate early restoration of knee ROM during the postoperative period.[17,18] In contrast, one recent retrospective study of 29,888 patients found that a 6 week or longer delay of ACLR reduces the 2-year likelihood of surgical treatment of arthrofibrosis by 65% in patients aged younger than 40 years.[19] Although the timing of ACL surgery may affect the likelihood of postoperative arthrofibrosis, existing evidence suggests that early surgery combined with accelerated rehabilitation can lead to earlier return of strength and an infrequent incidence of arthrofibrosis. Conversely, delayed surgery may increase the risk of quadriceps atrophy and negatively influence postoperative knee function, especially problematic for high-level athletes who aim for rapid return to preinjury activity levels.

COMPARISON OF CLINICAL OUTCOMES AFTER EARLY AND DELAYED ANTERIOR CRUCIATE LIGAMENT RECONSTRUCTION

During the years, surgical timing was determined as an influential factor on clinical outcomes after ACLR. Several studies compared the effects of early and delayed ACLR on the risk of graft failure and revision surgery. Economic and decision analysis using registry-based patient data from the Knee Anterior Cruciate Ligament Nonsurgical versus Surgical Treatment trial and Multicenter Orthopedic Outcomes Network cohort determined that early isolated ACLR results in less sick-leave days (on average −31.6 days) and superior gains in Quality Adjusted Life Years compared with rehabilitation with optional delayed ACLR. However, findings are conflicting regarding the association between ACLR timing and the risk of revision ACLR. One study of 11,867 patients reported an increased risk of revision ACLR with delayed ACLR (performed at least 6 months after injury) compared with early ACLR (performed within 2 months

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of injury).[20] Conversely, an increased revision risk (relative risk = 2.07) was reported in patients undergoing early isolated ACLR (defined as surgery performed within 3 months of injury).[21] Furthermore, a 2 to 3-fold revision ACLR rate was found in patients undergoing ACLR within 90 days of injury compared with patients undergoing surgery after 365 days from injury.[22] However, it is worth noting that the likelihood of revision after early and delayed ACLR is likely influenced by factors such as activity level after surgery and patient age, rather than the timing of surgery itself.[23] Therefore, the risk of revision should be considered multifactorial and should not solely rely on the timing of surgery.

COMPARISON OF FUNCTIONAL OUTCOMES AFTER EARLY AND DELAYED ANTERIOR CRUCIATE LIGAMENT RECONSTRUCTION

Another major controversy in ACL surgery is whether early or delayed ACLR leads to superior objective and subjective outcomes for patients. Several studies report comparable clinical and outcomes superior knee stability because of both early and delayed ACLR.[24,25] One systematic review found similar tibiofemoral laxity, assessed with the KT-1000 arthrometer, Lachman test, and the pivot shift test, in patients who underwent early ACLR (performed within 1 month of injury) versus those who underwent delayed ACLR (performed >8 weeks after injury).[26] No significant differences in postoperative knee ROM were found between the 2 groups.[26] Additionally, one recent study showed comparable patient-reported outcomes, with significant improvements in all subscales of the Knee Injury and Osteoarthritis Outcome Score (KOOS) and favorable return rate to preinjury Tegner activity level after both early and delayed ACLR.[17] Notably, a high proportion of patients in both the early and delayed ACLR groups were able to return to Tegner activity level 6 or higher, suggesting surgical timing may have a limited influence on the rate of return to sport.[17] Furthermore, objective International Knee Documentation Committee scores were shown to be similar among patient groups, with the majority of (82% vs 71%, respectively) of patients having grade AB pivot shift at the 12-month postoperative follow-up, indicative of restored knee laxity because of both early and delayed ACLR.[17]

Further research showed that patients with early, hamstring tendon ACLR achieve superior outcomes in terms of postoperative isokinetic knee flexion strength after 2 years compared with patients who underwent delayed ACLR.[27] Specifically, patients with early surgery achieved significantly greater improvement in flexor strength at 180° (8.1% improvement) and 240° (9.4% improvement) per second, without significant improvement in extensor strength.[27] In the same study, early ACLR was associated with fewer sick-leave days and decreased socioeconomic costs compared with delayed ACLR.[27] Consequently, while both early and delayed surgery seem to yield good to excellent functional outcomes, a combination of functional outcomes should be assessed to determine the optimal timing of isolated ACLR.

The Impact of Anterior Cruciate Ligament Reconstruction Timing on the Incidence of Secondary Intra-articular Injury

Due to conflicting results in the current literature, the influence of ACLR timing on the incidence of secondary intra-articular injury remains a topic of debate. Delayed surgery was proposed to lead to meniscal[8] and chondral[7] damage secondary to knee instability because of prolonged ACL deficiency.[28]

One retrospective study of 226 patients who underwent primary ACLR found that surgery delayed beyond 7 months of injury was associated with 4.1-fold odds of the presence of medial meniscus injury recoded at surgery compared with ACLR performed

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within 6 months of injury.[29] Furthermore, patients with primary ACLR beyond 5 months from injury had 1.9-fold odds for lateral meniscus injury compared with surgery performed within 4 months of injury.[29] A systematic review performed in 2018 found that 6 of the 7 included studies implicated the role of prolonged knee instability with delayed primary ACLR in the development of medial meniscus injury (odds ratio [OR] range 3.46–11.56), despite an inconclusive association of prolonged instability with lateral meniscus injury rate.[30] The more predominant role of chronic ACL deficiency on the prevalence of subsequent medial meniscus injury is further supported by additional recent studies from the literature, which also highlight inferior rates of repairability of these lesions with time.[23,31–34] A recent systematic review and meta-analysis concluded that primary ACLR delayed beyond 3 months of ACL injury results in a 2.24-fold odds of medical meniscus injury compared with surgery performed earlier than 3 months, whereas the causal effect of timing on the prevalence of meniscal injuries remains unclear.[8] Consequently, early ACLR may be recommended to protect the integrity of the medial meniscus from prolonged knee instability.

During the last 20 years, several studies corroborated the negative influence of delayed ACL surgery on chondral wear.[2,35,36] One systematic review of 40 articles found a positive correlation between the rate of cartilage injury and ACLR delayed beyond 12 months (risk ratio = 0.42).[31] Furthermore, retrospective analysis of 3976 patients undergoing isolated primary ACLR at a single institution determined that surgery performed 12 months from injury resulted in a 1.20 OR of cartilage injury rate.[32] Additionally, one United States integrated health-care registry study found that 24.8% of patients with early (<3 weeks from injury to surgery) ACLR presented with cartilage injury at the time of surgery, whereas patients with delayed (≥9 months) ACLR had a 40.0% rate of concurrent cartilage injury.[23] One recent systematic review and meta-analysis compared cartilage injury rates as a function of 3-month, 6-month, and 12-month time intervals from ACL injury to ACLR with ACLR performed within 3 months of injury.[7] The analysis of 14 studies included in the final synthesis found an increase of the prevalence of chondral injury at each time interval compared with early surgery (<3 months), with low-grade cartilage injuries becoming prevalent after the 3-month interval (OR = 1.91) and high-grade cartilage injuries becoming more prominent after 12 months (OR = 3.06).[7] In contrast, another recent meta-analysis found no difference in the rate of cartilage injury between early (<3 weeks) and delayed (>6 weeks) ACLR is prevalent in the setting of MLKI but not in isolated ligamentous injury only involving the ACL.[10] However, heterogeneity between different syntheses performed on the available literature is likely due to different interpretations of cutoff values for early and delayed surgery. Despite the difficulty to determine the causality between surgical timing and the development of intra-articular injuries due to the inability to obtain baseline MRIs at the time of injury, current evidence predominantly suggests that early ACLR has a protective effect on the development of further intra-articular pathology with time. Prevention of chondral damage in patients with ACL injury is essential because patients with concomitant partial-thickness and full-thickness cartilage lesions at the time of ACLR report inferior 5-year subjective knee function measured with the KOOS.[37]

INTERPLAY BETWEEN ANTERIOR CRUCIATE LIGAMENT RECONSTRUCTION TIMING AND CONCURRENT SURGICAL INTERVENTIONS

Injury to the ACL is frequently accompanied by concomitant injuries and adequate treatment consists of addressing concurrent pathologic condition with concurrent surgical interventions. However, the interplay between additional surgical interventions concurrent with ACLR and surgical timing remains to be clarified. Patients with

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high-grade rotatory knee instability may benefit from additional lateral extra-articular interventions concurrent to ACLR to restore rotatory knee stability.[38,39] One recent study compared postoperative outcomes among a cohort of patients with early (<8 weeks from injury to surgery) combined ACLR and anterolateral ligament (ALL) reconstruction and a cohort with delayed (>8 weeks from injury to surgery) combined ACLR and ALL reconstruction, with no differences in objective knee stability, subjective knee function, and graft failure rates (2.9% and 2.1%, respectively).[40] Furthermore, patients with ACL injury often present with concomitant injury to the medial collateral ligament (MCL).[4,41] Although the choice between surgical or nonoperative treatment of MCL injuries concomitant to ACLR remains a controversial topic, it was recently reported that as few as 10% of patients nonoperatively treated for MCL injury concurrent with ACLR return to preinjury activity level.[42] Due to the detrimental effect of persistent valgus laxity on the integrity of intra-articular soft-tissue, prolonged rehabilitation with incomplete healing and stress on the ACL graft, early combined ACLR and MCL reconstruction is increasingly favored.[43] Consequently, patients with anterolateral and anteromedial instability in the setting of surgically treated ACL injury may benefit from early ACLR with concurrent lateral extra-articular tenodesis and MCL reconstruction. Repair of treatable meniscus pathologic condition in conjunction with ACLR is also considered advantageous in patients with ACL injury. It was reported that 77.6% of meniscus injuries were repaired concomitant to early (<6 months from injury to surgery) ACLR, whereas only 46.7% of meniscus injuries were repaired with delayed (>6 months from injury to surgery) ACLR. Additionally, meniscus repair performed in the setting of early (<48 hours from injury) ACLR resulted in lower rates of extension deficit between 3° and 5° at the 12-month postoperative follow-up compared with delayed (performed in the inflammation-free period) ACLR and meniscus repair (3.7% vs 22.2%, respectively).[11] Therefore, early surgical timing for ACLR may be preferred in highly active patients with concurrent repairable meniscus injury to prevent the incidence of downstream chondral wear and posttraumatic OA.

SUMMARY

In light of the current evidence, an improved understanding of surgical timing is warranted to optimize patient outcomes with ACLR in the setting of isolated ACL and MLKIs. Although interpretation of the existing data is hampered by inconsistent definitions of surgical timing, early ACLR performed within weeks of injury may result in equivalent to superior clinical and functional outcomes. Importantly, early ACLR seems to be beneficial for decreasing the incidence of medial meniscus and chondral injury by avoiding prolonged knee instability, with excellent clinical and functional outcomes and no marked increase in the risk of postoperative arthrofibrosis compared with delayed surgery. Future studies are required to clarify the cutoff between early and delayed surgeries to determine the influence of concurrent surgical procedures on optimal surgical timing and to explore the role of surgical timing in the treatment of patients with MLKIs.

CLINICS CARE POINTS

- Delayed ACLR may increase the incidence of secondary intraarticular injuries and quadriceps atrophy, with a potentially negative impact on postoperative outcomes.
- Inconsistent definitions of early and delayed ACLR may lead to ambiguity and hinder evidence-based decision-making.

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- The timing of ACLR in the setting of MLKI can be challenging due to the complex nature of these injuries.
- Both early and delayed ACLR yield good to excellent functional outcomes, with patients often returning to preinjury activity levels.
- Patients with ACL injuries may benefit from concurrent surgical treatment for associated pathologies, such as lateral extraarticular tenodesis, MCL reconstruction, and meniscus repair.

DISCLOSURE

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